

Table of Laplace Transforms¹

$F(s)$	Number	$f(t)$
$\frac{1}{s}$	1	$u(t) = 1, t > 0$
$\frac{1}{s^2}$	2	$r(t) = t, t > 0$
$\frac{1}{s^n} (n = 1, 2, 3, \dots)$	3	$\frac{t^{n-1}}{(n-1)!}$
$\frac{1}{\sqrt{s}}$	4	$\frac{1}{\sqrt{\pi t}}$
$\frac{1}{s^{3/2}}$	5	$2\sqrt{\frac{t}{\pi}}$
$\frac{1}{s^\alpha} (\alpha > 0)$	6	$\frac{t^{\alpha-1}}{\Gamma(\alpha)}$
$\frac{1}{s+a}$	7	e^{-at}
$\frac{1}{(s+a)^2}$	8	te^{-at}
$\frac{1}{(s+a)(s+b)}$	9	$\frac{1}{a-b}(e^{-bt} - e^{-at})$
$\frac{1}{(s+a)^n} (n = 1, 2, \dots)$	10	$\frac{1}{(n-1)!} t^{n-1} e^{-at}$
$\frac{1}{(s+a)^\alpha} (\alpha > 0)$	11	$\frac{t^{\alpha-1} e^{-at}}{\Gamma(\alpha)}$
$\frac{s}{(s+a)(s+b)}$	12	$\frac{1}{b-a}(be^{-bt} - ae^{-at})$
$\frac{1}{s^2 + a^2}$	13	$\frac{1}{a} \sin at$
$\frac{s}{s^2 + a^2}$	14	$\cos at$
$\frac{1}{s^2 - a^2}$	15	$\frac{1}{a} \sinh at$
$\frac{s}{s^2 - a^2}$	16	$\cosh at$
$\frac{1}{s(s^2 + a^2)}$	17	$\frac{1}{a^2}(1 - \cos at)$

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Table of Laplace Transforms (*Continued*)

$F(s)$	Number	$f(t)$
$\frac{1}{s^2(s^2 + a^2)}$	18	$\frac{1}{a^3}(at - \sin at)$
$\frac{1}{(s^2 + a^2)^2}$	19	$\frac{1}{2a^3}(\sin at - at \cos at)$
$\frac{s}{(s^2 + a^2)^2}$	20	$\frac{t}{2a} \sin at$
$\frac{s^2}{(s^2 + a^2)^2}$	21	$\frac{1}{2a}(\sin at + at \cos at)$
$\frac{s^2 - a^2}{(s^2 + a^2)^2}$	22	$t \cos at$
$\frac{s}{(s^2 + a^2)(s^2 + b^2)}$	23	$\frac{\cos at - \cos bt}{b^2 - a^2} (a^2 \neq b^2)$
$\frac{1}{(s - a)^2 + b^2}$	24	$\frac{1}{b} e^{at} \sin bt$
$\frac{(s - a)}{(s - a)^2 + b^2}$	25	$e^{at} \cos bt$
$\frac{3a^2}{s^3 + a^3}$	26	$e^{-at} - e^{(1/2)at} \left[\cos\left(\frac{at\sqrt{3}}{2}\right) - \sqrt{3} \sin\left(\frac{at\sqrt{3}}{2}\right) \right]$
$\frac{4a^3}{s^4 + 4a^4}$	27	$\sin(at) \cosh(at) - \cos(at) \sinh(at)$
$\frac{s}{s^4 + 4a^4}$	28	$\frac{1}{2a^2} [\sin(at) \sinh(at)]$
$\frac{1}{s^4 - a^4}$	29	$\frac{1}{2a^3} [\sinh(at) - \sin(at)]$
$\frac{s}{s^4 - 8a^3s^2 + 4a^4}$	30	$\frac{1}{2a^2} [\cosh(at) - \cos(at)]$
$\frac{1}{(s^2 + a^2)^3}$	31	$(1 + a^2t^2) \sin(at) - at \cos(at)$
$\frac{1}{s} \left(\frac{s-1}{s} \right)^n$	32	$\frac{e^t}{n!} \frac{d^n}{dt^n} (t^n e^{-t})$ [n^{th} degree Laguerre polynomial]
$\frac{s}{(s - a)^{3/2}}$	33	$\frac{1}{\sqrt{\pi t}} e^{at} (1 + 2at)$
$\sqrt{s - a} - \sqrt{s - b}$	34	$\frac{1}{2\sqrt{\pi t^3}} (e^{bt} - e^{at})$
$\frac{1}{a + \sqrt{s}}$	35	$\frac{1}{\sqrt{\pi t}} - ae^{a^2t} \operatorname{erfc}(a\sqrt{t})$
$\frac{\sqrt{s}}{s - a^2}$	36	$\frac{1}{\sqrt{\pi t}} + ae^{a^2t} \operatorname{erf}(a\sqrt{t})$
$\frac{\sqrt{s}}{s + a^2}$	37	$\frac{1}{\sqrt{\pi t}} - \frac{2a}{\sqrt{\pi}} e^{-a^2t} \int_0^{a\sqrt{t}} e^{\beta^2} d\beta$

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Table of Laplace Transforms (Continued)

$F(s)$	Number	$f(t)$
$\frac{1}{\sqrt{s}(s-a^2)}$	38	$\frac{1}{a}e^{a^2t} \operatorname{erf}(a\sqrt{t})$
$\frac{1}{\sqrt{s}(s+a^2)}$	39	$\frac{2}{a\sqrt{\pi}}e^{-a^2t} \int_0^{a\sqrt{t}} e^{\beta^2} d\beta$
$\frac{1}{\sqrt{s}(a+\sqrt{s})}$	40	$e^{a^2t} \operatorname{erfc}(a\sqrt{t})$
$\frac{1}{(s+a)\sqrt{s+b}}$	41	$\frac{1}{\sqrt{b-a}}e^{-at} \operatorname{erf}[\sqrt{t(b-a)}]$
$\frac{\sqrt{s+2a}}{\sqrt{s}} - 1$	42	$ae^{-at}[I_1(at) + I_0(at)]$
$\frac{1}{\sqrt{(s+a)(s+b)}}$	43	$e^{-\frac{1}{2}(a+b)t} I_0\left(\frac{a-b}{2}t\right)$
$(\alpha > 0) \frac{\Gamma(\alpha)}{(s+a)^\alpha(s+b)^\alpha}$	44	$\sqrt{\pi} \left(\frac{t}{a-b}\right)^{\alpha-1/2} e^{-\frac{1}{2}(a+b)t}$ $I_{\alpha-1/2}\left(\frac{a-b}{2}t\right)$
$\frac{1}{(s+a)^{1/2}(s+b)^{3/2}}$	45	$te^{-\frac{1}{2}(a+b)t} \left[I_0\left(\frac{a-b}{2}t\right) + I_1\left(\frac{a-b}{2}t\right) \right]$
$\frac{\sqrt{s+2a}-\sqrt{s}}{\sqrt{s+2a}+\sqrt{s}}$	46	$\frac{1}{t}e^{-at} I_1(at)$
$\frac{1}{\sqrt{s^2+a^2}}$	47	$J_0(at)$
$\frac{[\sqrt{s^2+a^2}-s]^\alpha}{\sqrt{s^2+a^2}} (\alpha > -1)$	48	$a^\alpha J_\alpha(at)$
$\frac{1}{(s^2+a^2)^\alpha} (\alpha > 0)$	49	$\frac{\sqrt{\pi}}{\Gamma(\alpha)} \left(\frac{t}{2a}\right)^{\alpha-1/2} J_{\alpha-1/2}(at)$
$\frac{[\sqrt{s^2+a^2}-s]^\alpha}{\sqrt{s^2+a^2}} (\alpha > 0)$	50	$\frac{\alpha a^\alpha}{t} J_\alpha(at)$
$\frac{[s-\sqrt{s^2-a^2}]^\alpha}{\sqrt{s^2-a^2}} (\alpha > -1)$	51	$a^\alpha I_\alpha(at)$
$\frac{1}{(s^2-a^2)^\alpha} (\alpha > 0)$	52	$\frac{\sqrt{\pi}}{\Gamma(\alpha)} \left(\frac{t}{2a}\right)^{\alpha-1/2} I_{\alpha-1/2}(at)$
$\frac{1}{s} e^{-\alpha/s}$	53	$J_0(2\sqrt{\alpha t})$
$\frac{1}{\sqrt{s}} e^{-\alpha/s}$	54	$\frac{1}{\sqrt{\pi t}} \cos(2\sqrt{\alpha t})$
$\frac{1}{\sqrt{s}} e^{\alpha/s}$	55	$\frac{1}{\sqrt{\pi t}} \cosh(2\sqrt{\alpha t})$
$\frac{1}{s^{3/2}} e^{-\alpha/s}$	56	$\frac{1}{\sqrt{\pi \alpha}} \sin(2\sqrt{\alpha t})$

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Table of Laplace Transforms (*Continued*)

$F(s)$	Number	$f(t)$
$\frac{1}{s^{3/2}} e^{\alpha/s}$	57	$\frac{1}{\sqrt{\pi\alpha}} \sinh(2\sqrt{\alpha t})$
$\frac{1}{s^\mu} e^{-\alpha/s} (\alpha > 0)$	58	$\left(\frac{t}{\alpha}\right)^{(\mu-1)/2} J_{\mu-1}(2\sqrt{\alpha t})$
$\frac{1}{s^\mu} e^{\alpha/s} (\alpha > 0)$	59	$\left(\frac{t}{\alpha}\right)^{(\mu-1)/2} I_{\mu-1}(2\sqrt{\alpha t})$
$e^{-\alpha\sqrt{s}} (\alpha > 0)$	60	$\frac{\alpha}{2\sqrt{\pi t^3}} \exp\left(-\frac{\alpha^2}{4t}\right)$
$\frac{1}{s} e^{-\alpha\sqrt{s}} (\alpha \geq 0)$	61	$\operatorname{erfc}\left(\frac{\alpha}{2\sqrt{t}}\right)$
$\frac{1}{\sqrt{s}} e^{-\alpha\sqrt{s}} (\alpha \geq 0)$	62	$\frac{1}{\sqrt{\pi t}} \exp\left(-\frac{\alpha^2}{4t}\right)$
$s^{-3/2} e^{-\alpha\sqrt{s}} (\alpha \geq 0)$	63	$2\sqrt{\frac{t}{\pi}} \exp\left(-\frac{\alpha^2}{4t}\right) - \alpha \operatorname{erfc}\left(\frac{\alpha}{2\sqrt{t}}\right)$
$\frac{1}{s} \log s$	64	$\Gamma'(1) - \log t [\Gamma'(1) = -0.5772]$
$\frac{1}{s^k} \log s (k \geq 0)$	65	$t^{k-1} \left\{ \frac{\Gamma'(k)}{[\Gamma(k)]^2} - \frac{\log t}{\Gamma(k)} \right\}$
$\frac{\log s}{s-a} (a \geq 0)$	66	$e^{at} [\log a - Ei(-at)]$
$\frac{\log s}{s^2+1}$	67	$Si(t) \cdot \cos(t) - Ci(t) \cdot \sin(t)$
$\frac{s \log s}{s^2+1}$	68	$-Si(t) \cdot \sin(t) - Ci(t) \cdot \cos(t)$
$\frac{1}{s} \log(1+ks) (k > 0)$	69	$-Ei(-t/k)$
$\log\left(\frac{s-a}{s-b}\right)$	70	$\frac{1}{t}(e^{bt} - e^{at})$
$\frac{1}{s} \log(1+k^2s^2)$	71	$-2Ci(t/k)$
$\frac{1}{s} \log(s^2+a^2) (a > 0)$	72	$2 \log(a) - 2Ci(at)$
$\frac{1}{s^2} \log(s^2+a^2) (a > 0)$	73	$\frac{2}{a} [at \cdot \log(a) + \sin(at) - at \cdot Ci(t)]$
$\log\left(\frac{s^2+a^2}{s^2}\right)$	74	$\frac{2}{t} [1 - \cos(at)]$
$\log\left(\frac{s^2-a^2}{s^2}\right)$	75	$\frac{2}{t} [1 - \cosh(at)]$
$\arctan\left(\frac{k}{s}\right)$	76	$\frac{1}{t} \sin(kt)$
$\frac{1}{s} \arctan\left(\frac{k}{s}\right)$	77	$Si(kt)$
$e^{k^2s^2} \operatorname{erfc}(ks) (k > 0)$	78	$\frac{1}{k\sqrt{\pi}} \exp\left(-\frac{t^2}{4k^2}\right)$

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Table of Laplace Transforms (Continued)

$F(s)$	Number	$f(t)$
$\frac{1}{s} e^{k^2 s^2} \operatorname{erfc}(ks) (k > 0)$	79	$\operatorname{erf}\left(\frac{t}{2k}\right)$
$e^{ks} \operatorname{erfc}(\sqrt{ks}) (k > 0)$	80	$\frac{\sqrt{k}}{\pi \sqrt{t}} (t + k)$
$\frac{1}{\sqrt{s}} \operatorname{erfc}(\sqrt{ks})$	81	$\begin{cases} 0 & \text{when } 0 < t < k \\ (\pi t)^{-1/2} & \text{when } t > k \end{cases}$
$\frac{1}{\sqrt{s}} e^{ks} \operatorname{erfc}(\sqrt{ks}) (k > 0)$	82	$\frac{1}{\sqrt{\pi} (t + k)}$
$\operatorname{erf}\left(\frac{k}{\sqrt{s}}\right)$	83	$\frac{1}{\pi t} \sin(2k\sqrt{t})$
$\frac{1}{\sqrt{s}} e^{k^2/s} \operatorname{erfc}\left(\frac{k}{\sqrt{s}}\right)$	84	$\frac{1}{\sqrt{\pi t}} e^{-2k\sqrt{t}}$
$K_0(ks)$	85	$\begin{cases} 0 & \text{when } 0 < t < k \\ (t^2 - k^2)^{-1/2} & \text{when } t > k \end{cases}$
$K_0(k\sqrt{s})$	86	$\frac{1}{2t} \exp\left(-\frac{k^2}{4t}\right)$
$\frac{1}{s} e^{ks} K_1(k\sqrt{s})$	87	$\frac{1}{k} \sqrt{t(t + 2k)}$
$\frac{1}{\sqrt{s}} K_1(k\sqrt{s})$	88	$\frac{1}{k} \exp\left(-\frac{k^2}{4t}\right)$
$\frac{1}{\sqrt{s}} e^{k/s} K_0\left(\frac{k}{s}\right)$	89	$\frac{2}{\sqrt{\pi t}} K_0(2\sqrt{2kt})$
$\pi e^{-ks} I_0(ks)$	90	$\begin{cases} [t(2k - 1)]^{-1/2} & \text{when } 0 < t < 2k \\ 0 & \text{when } t > 2k \end{cases}$
$e^{-ks} I_1(ks)$	91	$\begin{cases} \frac{(k - t)}{\pi k \sqrt{t(2k - 1)}} & \text{when } 0 < t < 2k \\ 0 & \text{when } t > 2k \end{cases}$
$-e^{as} Ei(-as)$	92	$\frac{1}{t + a} (a > 0)$
$\frac{1}{a} + se^{as} Ei(-as)$	93	$\frac{1}{(t + a)^2} (a > 0)$

The exponential, cosine, and sine integral functions are defined as

$$Ei(t) = \int_{-\infty}^t \frac{e^{\beta}}{\beta} d\beta$$

$$Ci(t) = \int_{\infty}^t \frac{\cos \beta}{\beta} d\beta$$

$$Si(t) = \int_0^t \frac{\sin \beta}{\beta} d\beta$$